

# Motor Imagery based EEG Signal Classification using Multi-scale CNN Architecture

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**Abstract**—Deep learning has been employed in EEG signal classification in recent years since it has multiple advantages, including automatic feature extraction and classification. Although deep learning models have been used recently for motor imagery signal classification, still there are challenges to improve the performance of classification. One of the challenges is the small size of the training dataset. Also, most of the existing CNN models utilize the features extracted from the last layer, which might result in the loss of some specific details. A multi-scale CNN architecture is proposed in this work to utilize features at the initial and final layers for classification of MI signals. Firstly, the size of the dataset is increased by using the sliding window data augmentation technique and time-frequency images are obtained after applying short time Fourier transform (STFT). The images are then fed to the multi-scale CNN. A mean kappa value of 0.803 and a mean classification accuracy of 85.3 percent is achieved for this model on dataset 2a of BCI competition IV through subject-specific training.

**Index Terms**—Motor imagery, data augmentation, short time Fourier transform, multi-scale CNN.

## I. INTRODUCTION

Brain-Computer Interface (BCI) links the human brain with external devices. It collects and analyses brain signals before converting them into commands for the output device to carry out the desired operation. The acquisition of brain signals can be done using invasive or non-invasive techniques. In non-invasive technique, the brain signals are acquired from electrodes that are placed on the scalp. Magnetoencephalography (MEG), positron emission theory (PET), electroencephalography (EEG), are some of the methods using non-invasive techniques. EEG is commonly employed non-invasive BCI data acquisition technique because of its benefits such as cost-effectiveness and simplicity to use. EEG signals are generally acquired through electrodes that measure the electric potential difference between two points. Motor imagery (MI) signal classification is one among the prominent research areas in EEG based BCI application because of its advantages in various fields like, medicine, sports, music etc. Motor imagery indicates a person thinks about executing a task without doing it. In BCI systems for motor-imagery applications, the main two steps in analyzing brain signals are, feature extraction and classification. Power spectral density estimation, common spatial pattern (CSP), filter bank CSP are some techniques used to get discriminate features from the EEG signals. STFT,

fast Fourier transform, continuous wavelet transform, wavelet packet decomposition are some methods used to extract time and frequency features. In [1] [2] [3] STFT, CWT methods are used in order to convert the data of 1-D time-series to 2-D time-frequency images. In [4] [5], the data augmentation technique increases the samples, and STFT is used to get time-frequency images from time signals. Bayesian classifier, artificial neural network (ANN), k-nearest neighbour, support vector machine (SVM) are some of the classification algorithms.

Deep learning models have been excelling in image classification and speech recognition from the past few years because of its ability to perform automatic feature extraction and classification. Deep learning has had a significant impact in medical research in recent years. In [6], shallow convolution network, which consists of 2 convolution layers and one dense layer, is used for extraction of features and classification of MI-EEG signals. In [7], shallow CNN with two convolution layers is used for classification. In [1] [3], the time-frequency images are given to deep CNN models for classification, which extract features at the last layers for classification. In [2], pre-trained CNN models Alex net, Resnet50, InceptionV3 are used to classify time-frequency images. In [8], the authors designed a multi-level feature fusion CNN model that extracts features at multiple-level layers corresponding to local and global input information. In addition, they achieved better performance of classification accuracy in comparison with the CNN model. In [4], multi-scale CNN with two convolution layers is used, which extracts features at both initial and last level layers for MI-EEG signal classification. Motivated by the above literature, this paper proposes a multi-scale convolution neural network model to get better classification results. In this experiment, data is given to the data augmentation technique to increase the size of dataset. Then the signals are converted to images using STFT, and the images are provided as input to multi-scale CNN for classification. The outline of this paper is represented in this sequence: Section II explains the methodology, data set description, data augmentation technique, data pre-processing using STFT and proposed multi-scale CNN architecture. Section III describes the experimental results. Lastly, section IV express the conclusion of the work.

## II. METHODOLOGY

The methodology followed in this paper for the motor imagery EEG signal classification is demonstrated in Fig. 1. The motor imagery EEG signals are taken from 9 subjects and are augmented to increase the training sample size which will further improve the model efficiency. Then the time series MI-EEG signals are pre processed by transforming them into time-frequency images using STFT. These images are provided as input to multi-scale CNN model for feature extraction and classification.

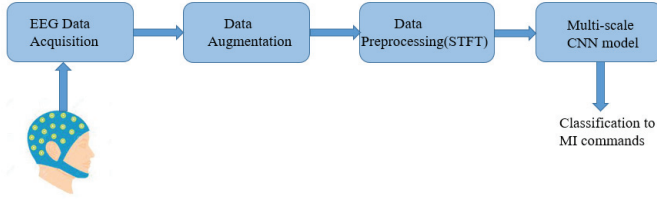


Fig. 1. Methodology for motor imagery EEG signal classification.

### A. Dataset Description

Dataset 2a of BCI competition IV was chosen for this experiment, which has 4 classes [9]. The MI-EEG signals are taken from 9 participants (subjects). The subjects are seated in front of the monitor in an armchair and are asked to imagine the movement of motor imagery action when the arrow-shaped cue is appears pointing right, left, up or down, it denotes one of the four classes. The four motor imagery actions are right hand (class1), left hand (class2), both feet (class3) and tongue (class4). The data is taken from 25 electrodes (channels), out of which 3 are EOG channels and 22 are EEG channels. The sampling frequency (fs) of collected data is 250 Hz and is bandpassed between 0.5 and 100 Hz. The data is recorded for two sessions, with each session consisting of 288 trails. Each trail is considered for 6 seconds. At starting of each trail, a fixation cross appears on the monitor for the subject to focus on screen. After 2 seconds a cue appears on the monitor and stays for 1.25 seconds which prompts the participant to imagine desired motor imagery action. The subjects were asked to perform the imagination of motor action until the disappearance of fixation cross. After a short break the monitor screen turns black again. This timing scheme repeats for all trails.

### B. Data Augmentation

Data augmentation technique is performed to increase the similar signals in the dataset, which helps to train the model more efficiently. In this paper, the data augmentation technique used is sliding window. Since the window length should be less than the signal length, the window length is taken for 4 seconds (1000 samples), with slide 0.1 seconds, as shown in Fig. 2. Finally, 5 similar signals for each signal are obtained after data augmentation.

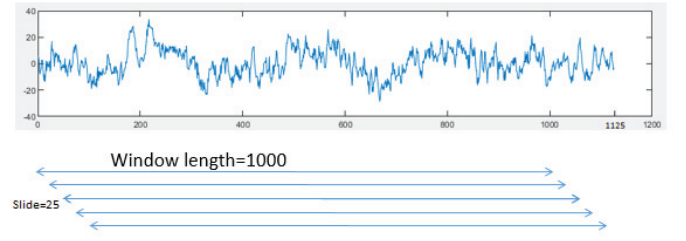


Fig. 2. Data augmentation using sliding window technique.

### C. Data Pre-processing

In this experiment, the data pre-processing technique which converts the time series input data to time-frequency images is short time Fourier transform method. STFT is a Fourier technique in which the signal is multiplied with a window, forming segments of the signal, and the Fourier transform is performed on these segments separately, as shown in (1).

$$A(k, \omega) = \int_{-\infty}^{\infty} a(t)w(t-k)e^{-i\omega t} dt \quad (1)$$

where  $a(t)$  is the time-series EEG signal,  $\omega(t-k)$  is the window multiplied with raw signal forming a segment of raw signal. With the variation of  $k$  value, the window shifts along with the signal, forming different segments. The Fourier transform is applied to those segments and forms a time-frequency spectrum. The window size is taken as 64 with an

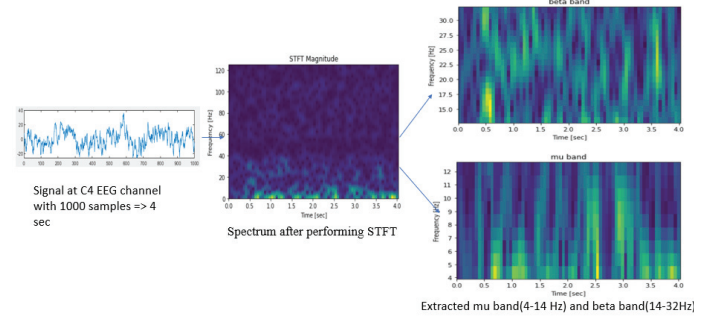


Fig. 3. Conversion of time to time-frequency spectrum.

overlap of 50, which gives the time-frequency spectrum for over 4 seconds (73-time bins) and 125 Hz (fs/2) frequency. The event-related synchronisation (ERS) / event-related desynchronization (ERD) phenomenon occurs in the  $\mu$  band (8 to 13 Hz) and  $\beta$  band (13 to 30 Hz) during motor imagery action. The energy in the brain's motor cortex region decreases at  $\mu$  band frequencies. This phenomenon is called ERD and the phenomenon where the energy increases at  $\beta$  band frequencies is called ERS. In this experiment, we extract the  $\mu$  and  $\beta$  bands of frequencies 4 to 14 Hz and 14 to 32 Hz, corresponding to ERD and ERS phenomena from the spectrums obtained after STFT, as shown in Fig. 3. The  $\mu$  band image is of size 10 x 73, and the  $\beta$  band is of size 18 x 73. The  $\beta$  band image is resized to 10 x 73 such that both the  $\mu$  and  $\beta$  bands will

have the same effect in classification [4]. The image obtained on stacking  $\mu$  and  $\beta$  band images is of size 20 x 73. The EEG signals from the 22 channels are transformed to time-frequency images. When the  $\mu$  and  $\beta$  of all the EEG channels are stacked, the obtained image's size is 440 x 73 for one class.

#### D. Proposed Multi-scale CNN Architecture

Generally basic CNN model consists of a convolution layer, pooling layer, flatten layer and an output layer. In the convolution layer, the feature maps are generated when the input is multiplied by the weights in filters. The pooling layer down-samples the feature maps, which reduces the dimensionality without losing important information. Flatten layer creates a single long feature vector from the feature maps obtained after pooling. In the output layer, the features extracted are classified into desired classes.

The proposed multi-scale CNN model in this paper consists of eleven layers: one input layer, three convolution layers, three max-pooling layers, two flatten layers, one concatenate layer and an output layer with a softmax activation function, as shown in Fig. 4. The configuration of network with number of filters used for each layer and the size of filters is shown in table I.

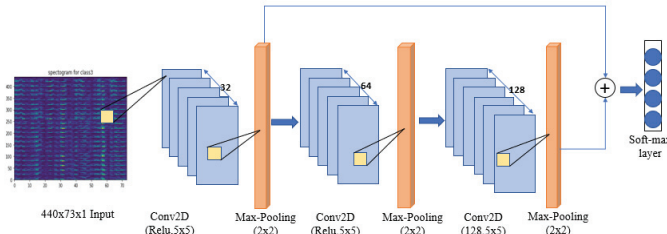


Fig. 4. The proposed multi-scale CNN model.

The 440 x 73 x 1 image generated after the EEG signals pre-processed by STFT are given to the input layer. In convolution layer, a set of convolution kernels are convolved with the input image, which gives two-dimensional feature maps. The activation function used after convolution is RELU, to reduce the linearity caused due to convolution operation. The feature maps are then given to the max-pooling layer, which reduces the network parameters and dimensionality of feature maps. After flattening the feature maps obtained from one convolution layer and max-pooling layer, initial level features are obtained. After three convolution layers and max-pooling layers, the features are flattened, and final level features are obtained. In many of the existing models, only last level features are used for classification, which may lead to loss of some detailed information. In this work, initial level and final level features are concatenated to form multi-scale features. The features are then connected to output dense layer

with softmax activation function which gives probabilistic distribution of predicted classes as in (2).

$$\sigma(y)_i = \frac{e^{y_i}}{\sum_{j=1}^T e^{y_j}} \quad (2)$$

Where  $\sigma(y_i)$  is the normalized probability for all the classes of an input,  $y_i$  is the original output for a particular class, T is number of classes. In this model, sparse categorical cross-entropy loss function is used. During model training, the hyper parameters are updated in order to minimise the loss function. The configuration for the proposed model is shown in table I.

### III. EXPERIMENTAL RESULTS

Dataset 2a of BCI competition IV for one session consisting of 288 trials for each subject is taken in this experiment. Out of 25 channels, only 22 EEG channels are required for classification, so the 3 EOG channels are removed. In the timing scheme of trail, from 3 to 6 seconds, motor imagery action is performed. So, to cover the entire motor imagery action, the data is taken from 2 to 6.5 seconds (4.5) which is 4.5 x 250 = 1125 samples. So, the length of the signal is 1125. Therefore, the shape of the data formed is 288 x 1125 x 22 (trials x no. of samples x no. of channels). After data augmentation using a sliding window with window length of 4 seconds (1000 samples) and slide of 0.1 seconds, we get 5 similar samples for each trail, so the number of samples increased to 288 x 5 = 1440. After data augmentation, the shape of the data is 1440 x 1000 x 22. Then the signals are pre-processed using STFT and form the spectrum size 440 x 73 for each signal. So, the data after pre-processing is of shape 1440 x 440 x 73 x 1. For each subject, out of 1440 samples, 80 percent is taken for training and 20 percent for testing the model. The dataset is tested using CNN model without concatenating features and the proposed multi-scale CNN model based on classification accuracy and kappa value. Table II shows the subject specific classification accuracy and kappa value.

#### A. Classification Accuracy

The average accuracy of classification achieved after testing the proposed model and CNN model is 85.3 percent and 73.39 percent, respectively. The subjects S1 attains the highest accuracy and S9 attains the least accuracy among all the subjects as in table II. The classification accuracy is compared with the previous models CNN, multi-CNN fusion, EEGNet, Shallow CNN, which uses the same dataset as shown in table III.

#### B. Kappa value

Cohen's kappa value is a statistical indicator of inter-rater reliability regarding classification accuracy. It means that the classification accuracy is compared with the random accuracy, and it removes the error produced by random accuracy. The kappa value is calculated as shown in (3). The average kappa value achieved by the multi-scale CNN model is 0.803.

$$\kappa = \frac{\text{classification accuracy} - \text{rand accuracy}}{1 - \text{rand accuracy}} \quad (3)$$

TABLE I  
CONFIGURATION OF NETWORK

| Layer            | Filters | Size of filter | Connected to     | Output shape |
|------------------|---------|----------------|------------------|--------------|
| InputLayer       | -       | -              | -                | (440,73,1)   |
| Conv2D I         | 32      | [5 x 5]        | InputLayer       | (440,73,32)  |
| MaxPooling2D I   | -       | [2 x 2]        | Conv2D I         | (220,36,32)  |
| Conv2D II        | 64      | [5 x 5]        | MaxPooling2D I   | (220,36,64)  |
| MaxPooling2D II  | -       | [2 x 2]        | Conv2D II        | (110,18,64)  |
| Conv2D III       | 128     | [5 x 5]        | MaxPooling2D II  | (110,18,128) |
| MaxPooling2D III | -       | [2 x 2]        | Conv2D III       | (55,9,128)   |
| Flatten I        | -       | -              | MaxPooling2D I   | (253440)     |
| Flatten II       | -       | -              | MaxPooling2D III | (63360)      |
| Concatenate      | -       | -              | Flatten I        | (316800)     |
|                  |         |                | Flatten II       |              |
| Dense            | -       | -              | Concatenate      | 4            |

TABLE II  
CLASSIFICATION ACCURACY FOR STFT+CNN, STFT+MULTI-SCALE CNN ON DATASET 2A OF BCI COMPETITION IV.

| Subjects      | STFT + CNN model |             | STFT+multi-scale CNN |             |
|---------------|------------------|-------------|----------------------|-------------|
|               | Accuracy(%)      | kappa value | Accuracy(%)          | kappa value |
| S1            | 75.9             | 0.678       | 96.88                | 0.958       |
| S2            | 63.1             | 0.297       | 88.61                | 0.848       |
| S3            | 74.5             | 0.66        | 82.99                | 0.773       |
| S4            | 58.91            | 0.452       | 79.86                | 0.731       |
| S5            | 74.75            | 0.663       | 76.04                | 0.68        |
| S6            | 64.85            | 0.531       | 86.72                | 0.822       |
| S7            | 82.18            | 0.7624      | 87.85                | 0.838       |
| S8            | 81.44            | 0.752       | 93.75                | 0.916       |
| S9            | 84.9             | 0.798       | 75.1                 | 0.668       |
| Mean Accuracy | 73.39            | 0.645       | 85.3                 | 0.803       |

TABLE III  
COMPARISION OF CLASSIFICATION ACCURACY OF SAME DATASET.

| Methods                                                             | Accuracy(%) |
|---------------------------------------------------------------------|-------------|
| SCNN [6]                                                            | 75.58       |
| Multi CNN Fusion [10]                                               | 74.8        |
| Alex net [11]                                                       | 81.09       |
| CNN [12]                                                            | 73.4        |
| EEGNet [13]                                                         | 69.0        |
| Multi-level feature fusion CNN [8]                                  | 74.5        |
| <b>Data Augmentation + STFT + Multi-scale CNN (proposed method)</b> | <b>85.3</b> |

Here the random accuracy is taken as 0.25 since the number of classes are 4. Kappa value indicates how well the model performed compared to how well it would have performed by chance.

For dataset 2a of BCI competition IV, the mean of classification accuracy obtained is 85.3 percent which is better than the existing models as shown in table III. And the average kappa value achieved is 0.803. Comparative analysis reveals that our proposed model has performed better.

#### IV. CONCLUSION

In this paper, data augmentation using a sliding window technique is applied on MI EEG signals to increase the

number of the signals for the model to perform better. Data augmentation is applied in such a way that the time which the subject imagines motor imagery action is covered. The  $\mu$  and  $\beta$  bands where the ERD and ERS phenomena occur are extracted for all 22 EEG channels from time-frequency image after pre-processing employing STFT. The proposed multi-scale CNN model concatenates features at initial and final level layers, which includes the detailed information useful for improving classification. Since the EEG data is less, the proposed model uses few number of convolution layers which reduces over fitting and improves classification of signals.

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